

F434 — F427 operator-level deepening (Cal #35 follow-up),
 posed as a sharp question, NOT over-claimed: the operator-level
 derivation of the Wolfenstein hierarchy reduces to ONE
 decidable question — does the substrate generation-overlap
 kernel compose MULTIPLICATIVELY ($V_{ub} \sim V_{us} \cdot V_{cb}$,
 observed) or ADDITIVELY ($V_{ub} \sim V_{us} + V_{cb}$, wrong)? I
 checked the clean candidate (geodesic-collinearity $\square$
 multiplicative) and found its premise FAILS, so I am NOT
 banking it. The candidate: IF the three generations were
 collinear on the dilation geodesic, geodesic distances would add
 ($d_{13} = d_{12} + d_{23}$) and coherent-state overlaps $\text{sech}^{\{2\nu\}}(d/2)$
 compose SUB-multiplicatively (verified: ratio < 1 , the right
 direction for the apex $|\rho - i\eta| \approx 0.4 < 1$) $\square$ Wolfenstein at operator
 level. BUT F384 establishes the τ is ρ -ROTATED ($u_\tau = \hat{\rho}$), NOT
 collinear with the μ ($u_\mu = \hat{e}_1$) — the generations sit at ANGLES
 in the rank-2 Cartan plane, not on one geodesic. So the
 collinearity premise is false and the clean
 geodesic-multiplicative derivation does NOT directly hold
 (caught my own over-eager idea via F384, Cal #27 at
 excitement). The angled picture naively gives ADDITIVE
 small-angle mixing ($V_{ub} \sim V_{us} + V_{cb}$, WRONG) unless the
 kernel is exponential/log-additive in the angle. SO the sharp
 operator-level question is: does Grace's rank-2
 generation-overlap kernel $\langle z|w \rangle = [N(z)N(w)]^{\{n_C/2\}/N(z,w)} \{n_C\}$,
 evaluated at the angled generation positions (μ at $\hat{e}_1$, τ at $\hat{\rho}$, e at
 its address), compose MULTIPLICATIVELY in the
 inter-generation separations? Multiplicative $\square$ Wolfenstein
 DERIVED at operator level (F427 upgraded principle $\rightarrow$ operator,

Cal #35 closed). Additive $\mathbb{R}$ tridiagonality is an empirical input, not substrate-forced (F427 stays principle-level). This is GRACE's geometry lane (the kernel composition law); I've reduced the Cal #35 operator-level question to it precisely. DISCIPLINE: did NOT bank the geodesic derivation (premise failed); did NOT fish the toy ratios (0.6-0.71, right ballpark for 0.4 but rank-1 toy, arbitrary distances). F427 Wolfenstein stays PRINCIPLE-level (Cal #35); operator-level reduces to Grace's kernel composition law. $m_e=R$ unaffected. Five-Absence passes. Count 9/26.

Lyra (Claude Opus 4.8)

2026-06-30 Tuesday (date-verified)

F434 — F427 operator-level: the decisive question is multiplicative-vs-additive kernel composition (Grace's lane)

Cal #35 flagged F427's Wolfenstein hierarchy at the principle level (adjacency), with the operator-level identification pending. Working it Tuesday morning — and the honest outcome is a sharply-posed question, not a forced derivation.

The candidate I checked (and why I am not banking it)

A clean operator-level mechanism would be: if the three generations are collinear on the dilation geodesic, then geodesic distances add ($d_{13} = d_{12} + d_{23}$), and the discrete-series coherent-state overlap $\text{sech}^{2\nu}(d/2)$ composes sub-multiplicatively (verified numerically: $\langle e|\tau\rangle/(\langle e|\mu\rangle\langle\mu|\tau\rangle) < 1$, since $\cosh(a+b) > \cosh a \cdot \cosh b$). That would give $V_{ub} = V_{us} \cdot V_{cb} \times (\text{correction} < 1)$ — the Wolfenstein hierarchy at operator level, with the correction in the right direction for the unitarity-triangle apex (observed $|\rho - i\eta| \approx 0.4 < 1$).

But the premise fails. F384 establishes that the τ is ρ -rotated (deposit direction $u_\tau = \hat{\rho}$), *not* collinear with the μ ($u_\mu = \hat{e}_1$) — the generations sit at angles in the rank-2 Cartan plane, not on one geodesic. So the collinearity assumption is false, and the clean geodesic-multiplicative derivation does not directly hold. (I caught my own over-eager idea against F384 — Cal #27 firing at the exciting step.)

The sharp operator-level question (handed to Grace's geometry)

With the generations at *angles* (not collinear), the naive small-angle reading gives additive mixing ($V_{ub} \sim V_{us} + V_{cb}$), which is wrong for the Wolfenstein hierarchy — *unless* the substrate kernel is exponential/log-additive in the angular separation. So the operator-level derivation reduces to one decidable question:

Does Grace's rank-2 generation-overlap kernel $\langle z|w \rangle = [N(z,z)N(w,w)]^{\{n_C/2\}/N(z,w)} \{n_C\}$, evaluated at the angled generation positions (μ at $\hat{e}_1$, τ at $\hat{\rho}$, e at its address), compose MULTIPLICATIVELY in the inter-generation separations?

- Multiplicative $\square$ Wolfenstein hierarchy derived at operator level $\rightarrow$ F427 upgraded principle $\rightarrow$ operator, Cal #35 closed, the apex correction (< 1) becomes a substrate prediction (magnitude possibly computable; phase stays INTERNAL, Cal #50).
- Additive $\square$ tridiagonality is an empirical input, not substrate-forced $\rightarrow$ F427 stays at principle level honestly.

This is Grace's geometry lane (the kernel composition law on the rank-2 positions); I have reduced the Cal #35 operator-level question to it precisely.

Discipline

- Did not bank the geodesic-collinear derivation — its premise (collinearity) is false per F384.
- Did not fish the toy ratios (0.60–0.71 for arbitrary rank-1 distances; right ballpark for the observed 0.4, but a rank-1 toy with hand-picked distances proves nothing).
- F427's Wolfenstein hierarchy stays at principle level (Cal #35); the operator-level derivation is open and reduces to Grace's kernel composition law.

m_e=R unaffected; no count move. Five-Absence passes. Count 9/26.